

Project title: Advanced Password Strength Analyzer and Custom Wordlist Generator

Introduction

Passwords continue to be one of the primary authentication mechanisms across digital systems. However, poor password practices make systems vulnerable to attacks. Understanding how password strength is calculated and how attackers construct custom wordlists is essential for cybersecurity awareness.

This project aims to:

- Analyse password strength using entropy-based calculations.
- Validate passwords against configurable security policies.
- Generate custom wordlists using user-specific data.
- Provide both CLI and GUI interaction.

Abstract

Password security remains a critical concern in modern cybersecurity environments. Weak passwords are commonly exploited using brute-force and dictionary-based attacks. This project presents a Python-based application designed to evaluate password strength using entropy calculations and configurable policy validation. Additionally, the tool generates realistic attack-style wordlists based on user-provided inputs using combinational leetspeak and permutation logic. The project demonstrates both defensive and offensive password security concepts through a user-friendly GUI interface.

Tools Used

- Python 3
- Tkinter for GUI development
- Regular Expressions for validation
- itertools for combinational logic
- JSON for policy configuration

Steps Involved in Building the Project

1. Designed a JSON-based password policy system.
2. Implemented regex-based validation checks.
3. Developed entropy calculation logic for strength measurement.

4. Created a scoring and classification mechanism.
5. Built a GUI interface using Tkinter.
6. Developed a modular wordlist generator.
7. Implemented combinational leetspeak transformations.
8. Added permutation logic and year appending.
9. Integrated file export functionality.
10. Tested and optimized functionality.

Conclusion

This project successfully integrates password strength analysis with realistic wordlist generation techniques. By combining entropy-based evaluation and advanced combinational transformations, the tool simulates real-world password attack methodologies while promoting secure password practices. The modular design and GUI integration enhance usability and demonstrate practical cybersecurity implementation skills.

Figure 1: Password Strength Checker GUI

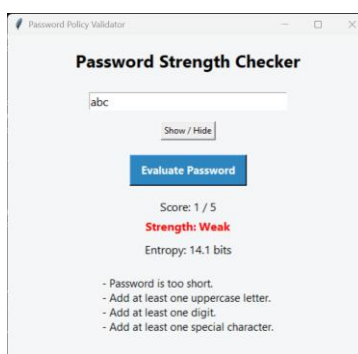


Figure 2: Custom Wordlist Generation Result

